VR Concept Definition

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1. Computer Simulation

A computer simulation is a virtual model that mimics how a real-world system/process behaves. Instead of testing things in reality (can be expensive/risky/impossible) we use computer to experiment safely (efficiently).

Examples
- Scientists simulate weather patterns to predict storms.
- Engineers simulate how a bridge will hold up before building it.
- Doctors use simulations to train for surgeries.

It's like creating a digital twin of something real, then watching how it reacts under different conditions. You can tweak variables, test "what if" scenarios, and learn without touching the actual thing.

1.1. Computer Simulation Pipeline

➔ **Define the System**
- Identify what you're simulating (e.g. traffic flow, weather, molecule behavior).
- Set goals: What do you want to learn or predict?

➔ **Build the Model**
- Create a mathematical or logical representation of the system.
- Choose variables, rules, and relationships.

➔ **Implement in Code**
- Translate the model into a computer program.
- Use simulation software or custom code (e.g. Python, MATLAB, Unity).

➔ **Run the Simulation**
- Input initial conditions.
- Let the system evolve over time or through iterations.

➔ **Analyze Results**
- Visualize outputs (graphs, animations, metrics).
- Compare with real-world data or expectations.

➔ **Refine & Repeat**
- Adjust parameters or model structure.
- Re-run to improve accuracy or explore new scenarios.

2. What is VR ?

Virtual reality is a computer-generated simulation that immerses the user in an artificial environment, **designed to be immersive, interactive, and multimedia**.

2.1. Virtual Simulation

The virtual world is **designed and animated** using: 3D modeling, Animations and Real-time graphic engines.

2.2. An Immersive Experience

VR **transfers the user's perception** from the real world to a virtual one. With headsets and sensors, it **engulfs the user** in a digital universe where the senses are stimulated.

2.3. Real-Time Interaction

Interaction is **bidirectional and synchronized**: The user acts on the virtual world, and it reacts instantly. To obtain interactivity, we need to define two scenarios:

- **Communication scenario –** defines how the user interacts with the virtual world.
- **Content scenario –** structures the actions, events, and responses of the virtual world.

2.4. A Multimedia Experience

VR engages **multiple sensory channels** to enhance immersion: Vision, Hearing and Touch (haptics).

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